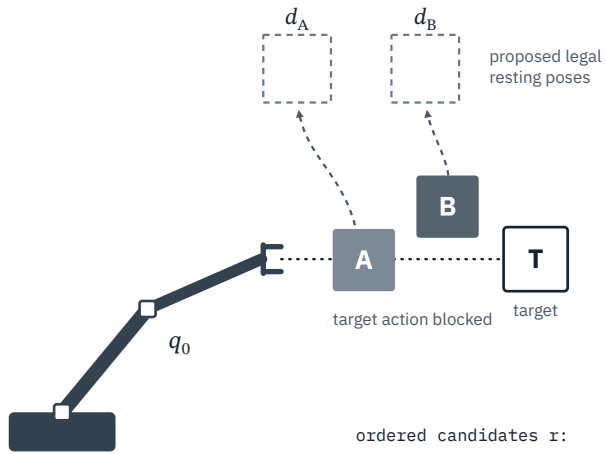
From infeasible manipulation to an ordered minimal repair

Learn to predict the consequences of high-level corrective actions **without planning every hypothetical intervention**.

ILLUSTRATIVE · NOT MEASURED



Initial scene, q_0, target → Candidate ordered object relocations r → Predict feasibility + select **no IK / planner** → Motion-plan and execute the selected repair

z_0	Does the target action succeed from the original scene at q_0?	no relocation; $z_R[\varnothing] \equiv z_0$
$z_S[I]$	Does the target succeed after a **hypothetical pose edit** I of objects, with the **robot still at** q_0?	fixed-robot, target-only query; does not show the robot can move them
$z_R[r]$	Does the full **physically ordered** repair, then the target action, then safe retreat succeed?	full motion + task teacher label
$z_O[r,t]$	Does the complete repair with action t removed succeed?	new physical sequence $r \setminus t$; omitted object stays put

$$r = [(o_1, d_1), \ldots, (o_m, d_m)], \quad x_t(r) = x_0 \text{ after the first } t \text{ proposed moves}$$

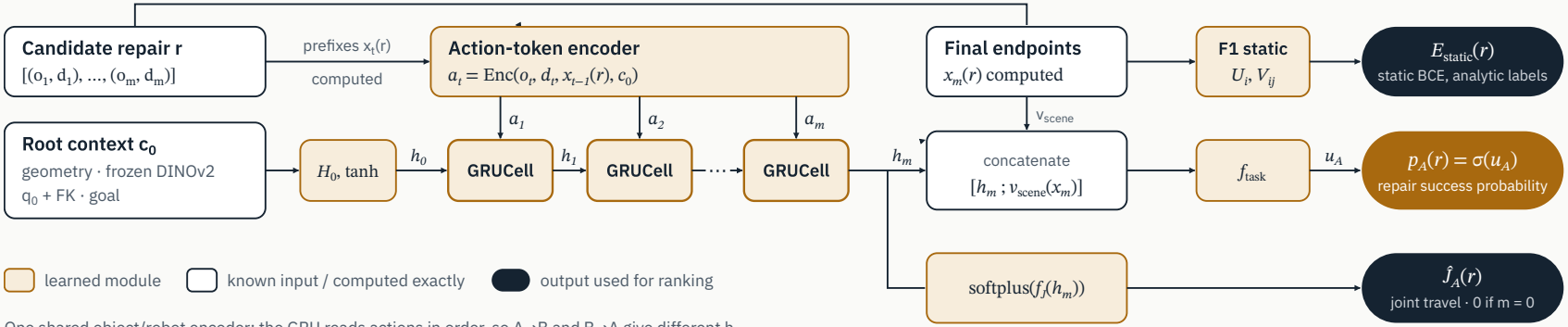
d_i is a legal resting pose, not a trajectory. Ideal prefix geometry is computed from the proposal, not observed from execution. A→B and B→A are different physical queries.

SHARED INPUTS initial RGB-D + masks → frozen mask-pooled DINOv2 · object shape · pose · relations · robot q_0 + FK · goal · candidate ordered (object, destination) sequence

Never inputs: planner/IK outcomes, executed trajectories, post-edit images, labels

A learns a repair success score directly from the ordered action sequence

GRU over ideal prefix-state action tokens, plus a learned local static energy. **No joint outcome factor graph.**



One shared object/robot encoder; the GRU reads actions in order, so A→B and B→A give different h

$$h_t = \mathrm{GRUCell}(a_t, h_{t-1})$$

$$E_{\mathrm{static}} = \sum_i U_i + \sum_{i<j} V_{ij}$$

$$E_{\mathrm{A}}^{\mathrm{rank}} = E_{\mathrm{static}} - T \log p_{\mathrm{A}}(r)$$

A's what-ifs = rescore, not a joint factor graph

SCENE EDIT I

Edit object geometry, q_0 unchanged → empty sequence → same f_{task} → p_I^{S}

OMISSION t

Remove action t, rebuild prefixes with that object unmoved → same GRU → p_t^{O}

Each outcome is a separate Bernoulli score. **No z-to-z coupling.**

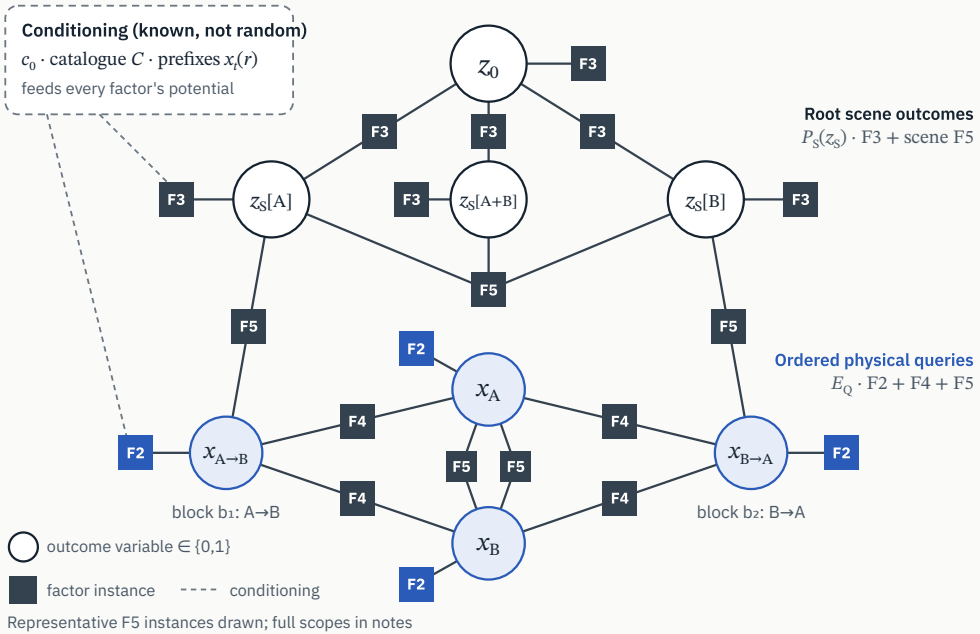
Training: factorized

Bernoulli NLL summed over **distinct observed physical queries** + static BCE + witnessed log-Huber J

Learns: geometry/appearance embeddings, local endpoint validity, sequence feasibility, edited-scene and omission predictions by rescoring, travel.

Does NOT learn: joint outcome couplings.

B assigns joint probabilities to combinations of original, edited, full-repair and omitted-repair outcomes



F1 · STATIC — OUTSIDE THE OUTCOME GRAPH

final poses $x_m(r)$ → F1 U_i F1 V_{ij} → $E_{static}(r)$

constant in z → cancels from P(z | r); trained by its own static BCE

F1	**Static** — unary U on endpoints, symmetric pair V on final poses; outside the outcome graph
F2	**Order** — one unary $-s_q x_q$ per distinct ordered query; predecessor/prefix pair pooling. **No GRU in B.**
F3	**Scene** — unaries on z_0 and fixed-q_0 edits $z_S[I]$; structured z_0–z_S pairs
F4	**Omission** — recomputed $r \setminus t$ shares the F2 scorer; structured complete ↔ omitted pair per block
F5	**Coupling** — scene–scene, scene–repair, omission–omission pairs; optional higher-order H (B-full)

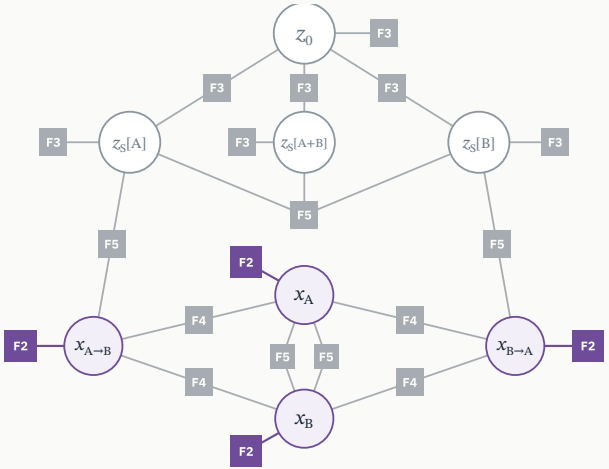
One binary variable + one sequence unary per distinct physical ordered query. A full repair and the same repair occurring as an omission are the SAME outcome variable: x_A is z_O[A→B, −B] and z_O[B→A, −B].

$$E_Q(x, z_S) = -\sum_{q \in Q} s_q x_q + \sum_{b \in \mathcal{B}} [E_{pair,b}(z_b) + E_{HO,b}(z_b)]$$

$$z_b = (z_S, x_{\sigma_b}),\ E_{pair,b} = -\tfrac{1}{2} z_b^\top W_b z_b,\ E_{HO,b} = -H_b(z_b)$$

$$P_B(z_S, x) = P_S(z_S) \frac{\exp[-E_Q(x, z_S)/T]}{\sum_{x'} \exp[-E_Q(x', z_S)/T]}$$

C holds the outcome graph fixed and tests whether GRU action memory improves order reasoning



Kept from B, unchanged in design: outcome variables z_0, z_S, x_q · F1 (outside) · F3, F4, F5 · conditional normalizer · joint masked NLL. **Purple = F2, the only replaced factor.** Weights trained separately.

B order	explicit predecessor interactions
C order	recurrent sequence memory
B & C	identical factor types and outcome ontology

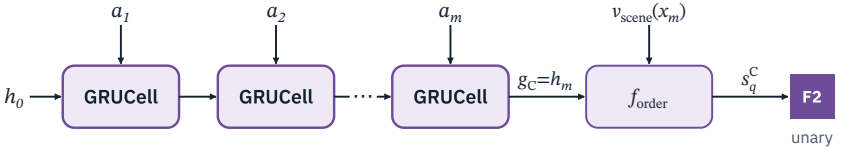
C is not "A + B". It is B's factor graph with an A-style GRU replacing B's order encoder.

B's F2 · predecessor interaction pooling — REMOVED IN C

$$b_t = \mathrm{MLP}_{\mathrm{mix}}([a_t\,;\,\mathrm{mean}_{u<t}\,\mathrm{MLP}_{\mathrm{pair}}([a_u\,;\,a_t])]),\quad g_B = \mathrm{mean}_t\, b_t$$

↓ **REPLACE — not add, not stack**

C's F2 · recurrent sequence memory



Omission $r \setminus t$: rebuild prefixes, run the GRU afresh → unary of the same alias variable $x_{r\setminus t}$

$$h_t = \mathrm{GRUCell}(a_t, h_{t-1}),\quad g_C = h_m,\quad s_q^C = f_{\mathrm{order}}([g_C\,;\,v_{\mathrm{scene}}(x_m)])$$

$$P_C(z_S, x) = P_S^C(z_S)\,\frac{\exp[-E_Q^C(x, z_S)/T]}{Z_Q^C(z_S)}$$

Training teaches outcomes; inference predicts interventions; execution uses the robot planner once committed

1 OFFLINE TEACHER · labels only

Target-only teleport check: edit I, robot fixed at $q_0 \to z_S[I]$

Full ordered repair + target + retreat, IK and motion planner allowed $\to x_q$. Different, independently tested scope.

VERIFIED	witnessed success → 1
CERTIFIED	sound scoped failure proof → 0
OPNEG-v1	budget negative → 0, policy-relative
UNKNOWN	undecided → marginalized
NOT_RUN	not queried → marginalized

2 LEARNED TRAINING

Build the root graph from c_0 and the catalogue; mask the observed bits.

B/C: **exact masked root NLL** by variable elimination; UNKNOWN and NOT_RUN bits are summed out.

Aliases appear once, as one variable and one label: $z_O[r,t] \equiv x_{r\setminus t}$, $z_R[\varnothing] \equiv z_0$

$E_{\text{static}}(r)$ is constant in z, so it cancels in $P(z \mid r)$ → separate static BCE. $\hat{J}$ → witnessed log-Huber.

A: factorized Bernoulli NLL on the same distinct decided queries.

3 INFERENCE → EXECUTION

New initial observation + catalogue → p_R, p^S, p^O with **zero IK / planner calls**.

Analytic static screens + calibrated admission threshold.

Lexicographic sort of admitted repairs (below).

Motion planner: precommit path feasibility of ranked fixed-order repairs → commit → execute once → single target attempt, no retry.

$$\mathcal{L}_{\text{B/C}} = \underbrace{-\frac{1}{N}\sum_c \log P_\theta(D_c)}_{\text{joint observed-outcome NLL}} + \underbrace{\lambda_s \mathcal{L}_{\text{static BCE}}}_{\text{F1, analytic labels}} + \underbrace{\lambda_J \mathcal{L}_{\text{logHuber}}(J)}_{\text{witnessed travel}}$$

$\lambda_s = \lambda_J = 1$. No additional κ loss or effect MSE by default; the goal is to learn through the joint outcome likelihood.

$$\text{rank}(r) = \text{lex}(\, m(r), \hat{J}(r), C_d(r), E_{\text{static}}(r) - T_s \log p_R(r) \,)$$

m move count and C_d object displacement are exact; $\hat{J}$ is learned joint travel. Deterministic ID breaks ties; r is not a random node of the outcome graph.

LEARNED θ · factor potentials s_q, W, H, scene prior, U, V, $\hat{J}$, encoders

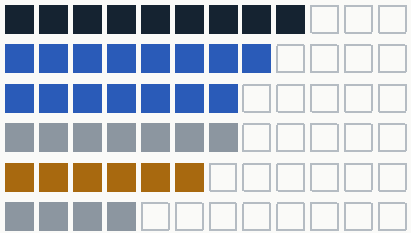
EXACT / ANALYTIC · prefixes $x_i(r)$, m, C_d, static screens and F1 labels, marginals

TEACHER (offline) · outcome labels, J witness; planner again only for real execution

Mechanism implemented; causal-repair advantage still must be proved

Historical P8d · executed 2D one-shot success

12 test roots · **old P8 checkpoints**, not the current unique-factor models · each cell = one root



No learned arm won a root that G+M lost. G+M's pose screens share the family that constructed these roots.

P9-L3 engineering · current unique-factor B/C

Exact inference matches brute force on 51 saved roots (max $|\Delta \log P| \approx 3.9 \cdot 10^{-14}$) · **375/375 tests**, 6/6 replays · 0 planner/IK calls in learned inference

No physical execution of the new unique-factor models yet.

Still missing scientific evidence

1	Proof-grounded complementarity $\varkappa = +1$ (current cases rest on a policy-relative OPNEG-v1 label)
2	Same-endpoint order swaps, physically decided from the same q_0
3	Independently decided omission necessity
4	Factor ablations on unseen scene-generation families
5	End-to-end executed benefit stronger than G+M

Caveats: few scenes and operational labels. P9-L3 validation has 12 roots, 3 of which carry ~95% of unique B-full NLL; unique vs PoE predictive parity is unresolved.

3D · PLANNED

2D object / mask / pose encoder → 3D RGB-D + multi-view · 6-DoF pose / shape / support · DINOv2 object features (Slot Attention optional) → unchanged A/B/C factor semantics: z_0, z_S, z_R, z_O, F1–F5 → 3D ground-truth teacher and execution

Can explicit learned intervention, omission and order couplings improve corrective TAMP beyond geometry-only screening and sequence scoring?